Supporting Information 2

2025-09-25

Data preparation

First, prepare data inputs for BioenergeticHSC batch method 1, which implements the NREI model. This will basically create a large grid of depth, velocity, and temperature combinations. Other parameters (e.g., fish size) could also be modified here if desired.

```
velocity <- seq(1,100, by=2)
depth <- seq(1, 100, by=2)
temp <- seq(5, 23, by=1)
hyd <- expand.grid(Velocity = velocity, Depth=depth)
hyd_full <- bind_rows(replicate(19, hyd, simplify = FALSE))
hyd_full$Temperature <- rep(temp, each = 2500)
hyd_full$Roughness <- ""
hyd_full$ForkLength <- 6.5 ## Fork length in cm
hyd_full$Mass <- 3 ## mass in g
hyd_full$Turbidity <- 0
hyd_full$DriftFile <-""
hyd_full$Notes <- ""
hyd_full$Label <- seq(1,47500, by=1)
Batch1Input <- hyd_full %>%
  dplyr::select(., Label, Depth, Velocity, Roughness, ForkLength, Mass,
                Temperature, Turbidity, DriftFile, Notes)
```

Functions

The following functions will be used on the output of BioenergeticHSC to convert energetics to a daily timestep. First, the function *dGEI* will compute a daily gross energy intake (J), which factors in the costs of non-foraging activity (assumed to be equal to SMR). Foraging duration is assumed to be 14 h, but this could be adjusted. The bioenergetics parameters are for coho, but could be adjusted for other species.

```
#### Individual functions for energetics parameters SMR (J/s)
base_smr <- function(temp, W) {
    SMR <- (1/3600) * exp(-2.94 + 0.87 * log(W) + 0.064 * temp)
    return(SMR * 14.1 * 1.75)
}


Cmax <- function(temp, W) {
    ## Cmax (g/g/day) - Water temperature and weight (W)
    CTO <- 15  # Temperature corresponding to 0.98 of Cmax
    CTM <- 18  # Temperature at which temperature dependence is 0.98 of max
    CTL <- 26  # Upper temperature
    CQ <- 5  # Lower water temperature
```

```r
    CK1 <- 0.42  # Unit-less fraction, see above
    CK4 <- 0.03  # Unit-less fraction, see above
    CA = 0.303  # intercept of the allometric function for a 1 g fish Topt
    CB = -0.275  # slope of allometric mass function
    # Temperature dependence function
    G1 <- (1/(CTO - CQ)) * log((0.98 * (1 - CK1))/(CK1 * 0.02))  #1st
    L1 <- exp(G1 * (temp - CQ))  #2nd
    Ka <- (CK1 * L1)/(1 + CK1 * (L1 - 1))  #3rd
    G2 <- (1/(CTL - CTM)) * log((0.98 * (1 - CK4))/(CK4 * 0.02))  #4th
    L2 <- exp(G2 * (CTL - temp))
    Kb <- (CK4 * L2)/(1 + CK4 * (L2 - 1))
    f_T <- Ka * Kb
    # Consumption equation. pCmax = 1, so drops out of equation
    Cmax <- (CA * W^CB) * f_T
    return(Cmax)
}

SwimCost <- function(temp, W, velocity) {
    ## Swimming cost (J/s)
    oq <- 14.1  ## Oxycaloric coefficient (Videler 1993)
    smr <- base_smr(temp, W)
    SC_chinook <- (1/3600) * oq * exp(-6.25 + (0.72 * log(W)) + (1.6 * log(velocity)))
    SC_coho <- (SC_chinook + (smr * 1.75))  ## Activity cost (J/s)
    return(SC_coho)
}

### Function that computes daily gross energy intake (J/day) less resting costs
### while not foraging
dGEI <- function(GREI, temp, W) {
    Cmax <- function(temp, W) {
        # Same Cmax function
        CTO <- 15
        CTM <- 18
        CTL <- 26
        CQ <- 5
        CK1 <- 0.42
        CK4 <- 0.03
        CA = 0.303
        CB = -0.275
        G1 <- (1/(CTO - CQ)) * log((0.98 * (1 - CK1))/(CK1 * 0.02))
        L1 <- exp(G1 * (temp - CQ))
        Ka <- (CK1 * L1)/(1 + CK1 * (L1 - 1))
        G2 <- (1/(CTL - CTM)) * log((0.98 * (1 - CK4))/(CK4 * 0.02))
        L2 <- exp(G2 * (CTL - temp))
        Kb <- (CK4 * L2)/(1 + CK4 * (L2 - 1))
        f_T <- Ka * Kb
        Cmax_i <- (CA * W^CB) * f_T
        return(Cmax_i)
    }
    ## Energy costs O2/h while resting for coho (Trudel and Welch 2005)
    SMR <- (exp(-2.94 + 0.87 * log(W) + 0.064 * temp)) * 1.75
    GREI_total <- GREI * 14 * 3600  ## GREI (J/s) - Assume 14 h foraging window
    ## Cmax in total J. 3626 J/g wet weight from Waters (1977).
```

```
    Cmax_J <- Cmax(temp, W) * 3626 * W
    ## Total resting costs (J), assuming fish are resting in zero velocity
    ## habitats.  SMR converted to J from Videler 1993
    Costs <- (SMR * 14.1) * 10
    ### Check if daily intake greater than Cmax, if it is then use Cmax
    DGEI <- ifelse(GREI_total < Cmax_J, GREI_total - Costs, Cmax_J - Costs)
    return(DGEI)
}
```

The input file is configured to predict habitat suitabiliy on a 2D depth and velocity grid. However, for visualization, univariate curves may be preferred. The following functions extract depth or velocity curves where the maximum for each is realized (i.e., a 1D vector that intersects the optimum of the 2D suitability matrix). In other words, each depth or velocity curve assumes that the other variable is not limiting suitability.

```
## Function to extract HSCs
### Extracts rows that correspond to optimal depths (for velocity HSC)
HSC_Vextract <- function(x){
  dd <- x$Depth[which.max(x$dHSI)]
  vhsc <- filter(x, Depth==dd)
  return(vhsc)
}

### Extracts rows that correspond to optimal depths (for depth HSC)
HSC_Dextract <- function(x){
  vv <- x$Velocity[which.max(x$dHSI)]
  dhsc <- filter(x, Velocity==vv)
  return(dhsc)
}
```

Data processing

```
##### Read in with batch method 1. Select and rename relevant columns
  bsc_input <- read.csv("./BioenergeticHSCbatch1output.csv")
  bsc <- na.omit(bsc_input)
  bsc_subset <- bsc %>%
    dplyr::select(., Depth..cm., Velocity..cm.s.,
                  Fork.length..cm.,Mass..g., Temperature,
                  Gross.rate.of.energy.intake..J.s.,
                        Total.energy.costs..J.s., )
  colnames(bsc_subset) <- c("Depth", "Velocity", "FL", "Mass",
                            "Temperature", "GREI", "TotalCosts")


## Compute daily gross energy intake per day with dGEI function
  bsc_subset$dGEI <-
    dGEI(bsc_subset$GREI, bsc_subset$Temperature, bsc_subset$Mass)
## Compute total daily activity costs (swimming and maneuvering while foraging)
### Ensure hours match dGEI function. 14 h assumed here
  bsc_subset$Costs_day <- bsc_subset$TotalCosts * 14 * 3600
### Daily net energy intake
```

```r
  bsc_subset$dNREI <- bsc_subset$dGEI - bsc_subset$Costs_day
  HSC_combined <- bsc_subset %>%
    dplyr::select(., GREI, Depth, Velocity, FL,
                                                 Mass, Temperature, TotalCosts,
                dGEI, Costs_day, dNREI)

## Proportion of Cmax
  HSC_combined$Cmax_J <-
    Cmax(HSC_combined$Temperature, HSC_combined$Mass) * 3626 * 3
  HSC_combined$pCmax <- HSC_combined$GREI*14*3600/HSC_combined$Cmax_J
  HSC_combined$pCmax <- if_else(HSC_combined$pCmax > 1, 1, HSC_combined$pCmax)

## Standardize to 0-1 scale.
##All negative daily NREI is set to 0.
##This data frame now contains temperature-dependent HSCs.
   HSC_combined$dHSI <- ifelse(HSC_combined$dNREI > 0,
                             HSC_combined$dNREI/max(HSC_combined$dNREI), 0)
   HSC_velocity <- HSC_Vextract(HSC_combined)
   HSC_depth <- HSC_Dextract(HSC_combined)
```

Plotting

Code to reproduce figures from the manuscript

```r
## Cmax across a range of temperature
  temp_range <- c(0:26)
  velocity <- c(0:50)
## Max consumption for 3 g fish across temperature range
  C_max <- Cmax(temp_range, 3)
## Make into data frame and convert to J
  cdat <- data.frame(Temperature = temp_range, Cmax=C_max*3626)
  cmax_plot <- cdat %>% ggplot(aes(x=Temperature, y=Cmax))+
    geom_line(lwd=1.25)+theme_bw()+ylab("Max Consumption (J/g/day)")


## SMR across a range of temperatures
  smr<- base_smr(temp_range, 3)
  smr_dat <- data.frame(SMR = smr, Temperature = temp_range)
  smr_plot <- smr_dat %>% ggplot(aes(x=Temperature, y=SMR))+
    geom_line(lwd=1.25)+theme_bw()+ylab("Standard Metabolic Rate (J/s)")

## Activity costs across velocity and temperatures
  temp_range_small <- c(6,12,18,24)
  tempxvelocity <- expand.grid(velocity, temp_range_small)
  names(tempxvelocity) <- c("Velocity", "Temperature")
  tempxvelocity$cost <-
    SwimCost(tempxvelocity$Temperature, 3, tempxvelocity$Velocity)
  tempxvelocity$Temperature <- factor(tempxvelocity$Temperature)
  activity_plot <-
    tempxvelocity %>% ggplot(aes(x=Velocity, y=cost, colour=Temperature))+
    geom_line(lwd=1.25)+theme_bw()+
    ylab("Activity Costs (J/s)") + xlab("Velocity (cm/s)") +
```

```
    scale_colour_manual(values=c("blue", "yellow","orange", "red"))

## Combine into one figure
#  fig1 <- grid.arrange(smr_plot, activity_plot,cmax_plot, nrow = 2,
#                  layout_matrix= rbind(c(1,2), 3))
# ggsave("./Figures/Fig1.pdf", fig1,
#            width = 10, height = 8, units = "in", dpi = 600)
```

Figure 2

```
## Illustrate modelling process and the influence of Cmax on daily NREI


  HSC_velocity$Cmax_12 <- Cmax(12, HSC_velocity$Mass)*3626*3
  HSC_velocity$Cmax_6 <- Cmax(6, HSC_velocity$Mass)*3626*3

## Plots of energy budgets against velocity at different temperatures

  example_6C <- HSC_velocity %>% filter(., Temperature == 6) %>% ggplot()+
    geom_line(aes(x=Velocity, y=dGEI), color="green", lwd=1.2)+
    geom_line(aes(x=Velocity, y=Costs_day), color="red", lwd=1.2) +
    geom_line(aes(x=Velocity, y=dNREI), color="black", lwd=1.2) +
    geom_line(aes(x=Velocity, y=Cmax_6), lty=2, lwd=1.2)+
    theme_bw() + ylim(0,2500)+ xlim(0,50)+ylab("Energy (J/day)") +
    xlab("Velocity (cm/s)")


  example_12C <- HSC_velocity %>% filter(., Temperature == 12) %>% ggplot()+
    geom_line(aes(x=Velocity, y=dGEI), color="green", lwd=1.2)+
    geom_line(aes(x=Velocity, y=Costs_day), color="red", lwd=1.2) +
    geom_line(aes(x=Velocity, y=dNREI), color="black", lwd=1.2) +
    geom_line(aes(x=Velocity, y=Cmax_12), lty=2, lwd=1.2)+
    theme_bw() + ylim(0,2500)+ xlim(0,50)+ylab("Energy (J/day)") +
    xlab("Velocity (cm/s)")

#fig2 <-  plot_grid(example_6C, example_12C,
#                   labels = c("6\u00b0 C", "12\u00b0 C"), label_x=0.5)
# ggsave("./Figures/Fig2.pdf", fig2,
#            width = 10, height = 8, units = "in", dpi = 600)
```

Figure 3

```
### Prepare data frames with univariate curves across temperatures
  HSC_velocity$Temperature <- factor(HSC_velocity$Temperature)
  HSC_depth$Temperature <- factor(HSC_depth$Temperature)

### Velocity plot
  vhsi_plot <- HSC_velocity %>%
    ggplot(aes(x=Velocity, y=dHSI, colour=Temperature))+
    geom_line(se=F, lwd=0.75)+theme_bw()+ylim(0,1)+
```

```r
    xlim(0,40)+xlab("Velocity (cm/s)") +
    ylab("Suitability (Standardized J/day)")+
    scale_colour_viridis(discrete = TRUE)+theme(legend.position = "none")

### Depth plot
  dhsi_plot <- HSC_depth %>% ggplot(aes(x=Depth, y=dHSI, colour=Temperature))+
    geom_smooth(se=F, lwd=0.75)+
    theme_bw()+ylim(0,1)+xlim(0,100)+xlab("Depth (cm)") +
    ylab(NULL)+ scale_colour_viridis(discrete = TRUE)+
    theme(axis.text.y=element_blank(),axis.ticks.y = element_blank())




#fig3 <- plot_grid(vhsi_plot, dhsi_plot +
#    guides(colour = guide_legend(override.aes = list(linewidth = 1.5))))
#   ggsave("./Figures/Fig3.pdf", fig3,
#           width = 10, height = 8, units = "in", dpi = 600)
```

Figure 4

```r
### Proportion of Cmax at the optimal depth/velocity across temperature
  pcmax <- HSC_velocity %>% group_by(Temperature) %>%
    filter(dHSI == max(dHSI)) %>% arrange(Temperature) %>%
    mutate(Temperature = as.numeric(Temperature))%>%
    mutate(ration = Cmax_J * pCmax)


   pcmax_plot <- pcmax %>% ggplot()+
    geom_hline(yintercept = 1, color="red", lty=2, lwd=2)+
    geom_point(aes((Temperature+4), pCmax,), size=3)+theme_bw()+
     ylab("Proportion of Cmax")+xlab("") + ylim(0.5,1)+
     theme(axis.text.x=element_blank(),axis.ticks.x = element_blank())



   ration_plot <- pcmax %>% ggplot()+
     geom_line(aes((Temperature+4), Cmax_J), colour="red", lwd=2) +
     geom_point(aes((Temperature+4), ration), size=3)+theme_bw()+
    ylab("Daily Ration (J)") + xlab("Temperature \u00b0C")

#  fig4 <- plot_grid(pcmax_plot, ration_plot, nrow = 2, align = "v")
#   ggsave("./Figures/Fig4.pdf", fig4,
#           width = 10, height = 8, units = "in", dpi = 600)
```